LAB REPORT: LAB 1 - REQUIREMENTS ELICITATION & MODELING

(Kyte)

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1. Cover Page

Title: Elicitation and Modeling Requirements for the Kyte Project Management Tool

Course Info: Software Architecture

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2. Abstract/Summary

The Kyte project aims to develop a streamlined collaborative task management tool based on the Kanban methodology.

This initial lab focused on requirements elicitation and modeling. We documented essential Functional, Non-Functional, and Architecturally Significant Requirements (ASRs) in a structured format.

The core system behavior was formally modeled using a UML Use Case Diagram, clearly defining system boundaries, external actors, and key behavioral relationships.

This foundation establishes the scope and the critical architectural drivers that will be addressed in subsequent labs, particularly the design of the Layered Architecture in next Lab 2.

3. Lab Specific Section: I. Requirements Elicitation & Modeling

This section documents the deliverables for Lab 1, as specified in the report requirements.

3.1 Software Requirements Specifications (SRS)

The following tables document the elicited requirements, which collectively form the basis for the Kyte design.

3.1.1 Functional Requirements (FRs)

These requirements define the specific behaviors and functions the Kyte system must provide.

ID	Description	Priority
FR-01	The system must allow a User to create new Boards and manage Columns (columns) within those boards (e.g., To Do, In Progress, Done).	High
FR-02	The system must allow Users to create Tasks within columns and edit their details (title, label, attachment, description, due date, assign members, checklist).	High

FR-03	The system must support drag-and-drop functionality to move Tasks between Columns to update their status.	Critical
FR-04	The system must allow a Workspace Owner / Admin to invite other users to join a board via email or username.	Medium
FR-05	The system must allow Users to add comments to a specific Task for collaboration.	Medium
FR-06	The system must allow Admin to change the visibility of the board (limit to team or public so other users on platform can see).	Medium

3.1.2 Non-Functional Requirements (NFRs)

These requirements specify criteria that can be used to judge the operation of the system, not specific functions.

ID	Attribute	Description	Impact
NFR-01	Performance (Response Time)	API response time for CRUD operations on Tasks and Columns must be < 100ms under normal load conditions (50 concurrent users).	High
NFR-02	Security (Access Control)	Only users who are explicitly added as members of a Board can view or edit the tasks within that board. All requests must be authenticated and authorized on the server-side.	Critical
NFR-03	Reliability (Data Integrity)	The system must ensure that simultaneous moves of the same task by different users do not result in data loss or corruption, using optimistic locking or conflict resolution mechanisms.	Critical
NFR-04	Availability	The system must maintain a minimum uptime of 99% during business hours (8 AM - 6 PM).	Medium
NFR-05	Scalability	The system must be capable of supporting horizontal scaling to handle growth from 100 to 1,000 concurrent active boards without degrading response time.	High
NFR-06	Usability (Responsiveness)	The user interface must be responsive and function correctly across desktop and mobile.	Medium

3.1.3 Architecturally Significant Requirements (ASRs)

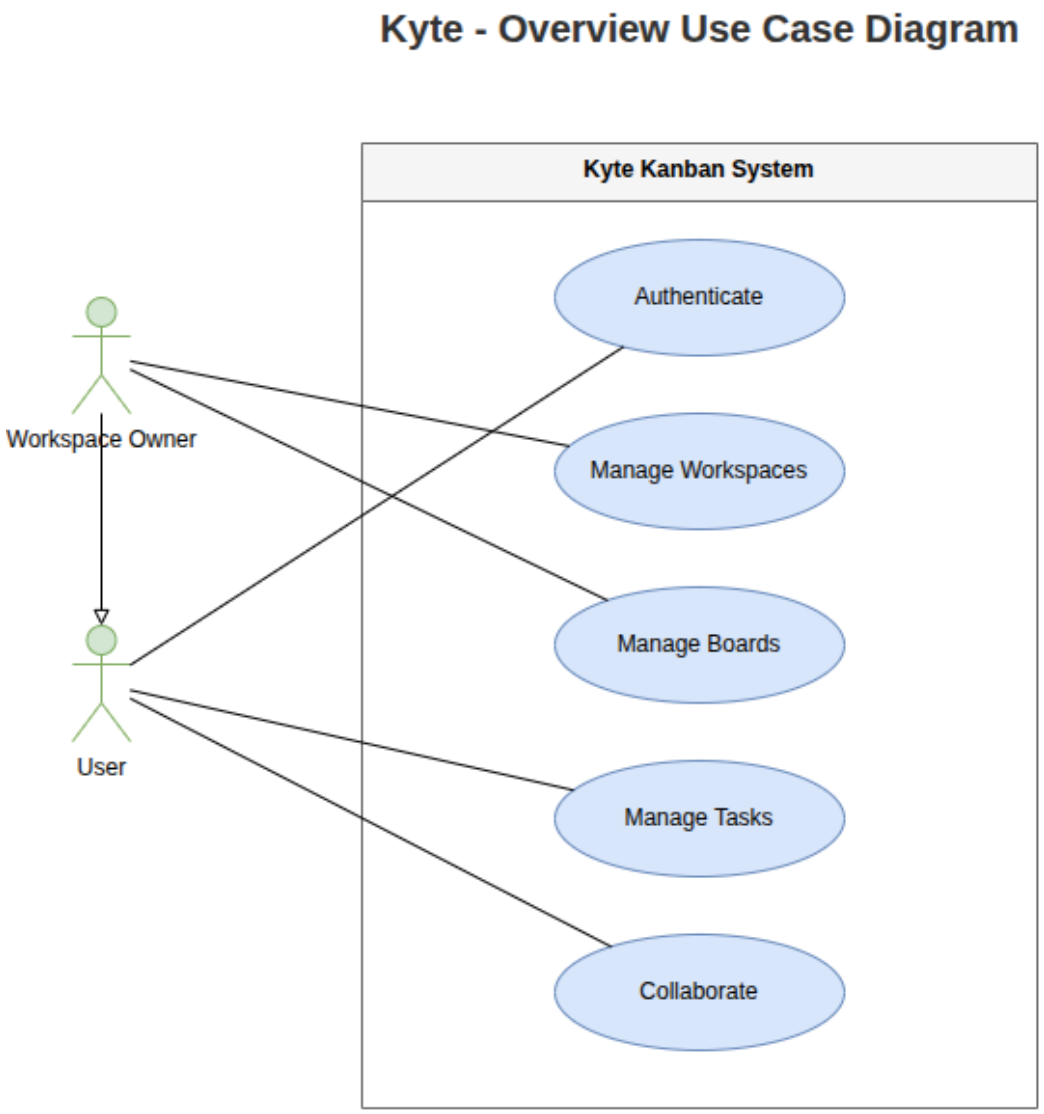
These are the non-functional requirements that have the most profound influence on the system's architecture, driving the selection of patterns.

ASR ID	Quality Attribute	Requirement Statement	Architectural Rationale
ASR-1	Modifiability	The core Business Logic must be strictly decoupled from the Web Interface so that a future Mobile Application (iOS/Android) can be developed using the same backend services without modifying existing code.	Enforces the principle of Separation of Concerns. Drives the design toward a layered structure where the Presentation Layer is physically separate from the Business Layer, allowing multiple interface types to share the same logic.
ASR-2	Security	All write operations (Create, Update, Delete) on Tasks and Columns must be validated against the user's Board Membership permissions on the server-side.	Necessitates a dedicated Security middleware to enforce access control at the Business Logic layer, protecting data regardless of the UI used.
ASR-3	Reliability	The system must implement optimistic locking or conflict resolution to handle concurrent task modifications without data loss.	Requires the Persistence Layer to implement versioning or timestamps on task entities, and the Business Logic Layer to handle conflict detection and resolution.
ASR-4	Scalability	The system must support stateless API design to enable horizontal scaling from 100 to 1,000 concurrent active boards.	Suggests avoiding stateful sessions in the web server and encourages RESTful API design within the Layered Architecture to facilitate future load balancing.
ASR-5	Performance	API endpoints must respond within 100ms for standard CRUD operations under	Drives efficient database query design, proper indexing in the Persistence Layer, and potential

		normal load (50 concurrent users).	caching strategies at the Business Logic Layer.

3.2 Modeling Artifact: UML Use Case Diagram

The UML Use Case Diagram models the functional scope of the Kyte system, showing the system boundary, the external Actors, and the core Use Cases.



- Actors: The diagram clearly identifies the primary actors: Board Member (Standard User) and Workspace Owner / Admin.
- System Boundary: The box delineates the scope of the Kyte application.
- Core Use Cases: Key high-level functions, such as Manage Board, Move Task, and Edit Task Details, are shown.

- Critical Path Flow: The diagram details the Move Task (Task Progression) process, which is the core value proposition of the system.
- Relationships Flow:
  - Include Relationship: The 'Validate Permissions' Use Case is a mandatory part of the 'Move Task' process, modeled using the <> relationship.
  - Extend Relationship: The 'Add Comment' Use Case is an optional behavior that extends the 'View Task Details' process, modeled using the <> relationship.

4. Architectural Design (note: Problem analysis of next lab)

The requirements elicited in Lab 1 have a direct impact on the design challenge of Lab 2: Layered Architecture Design.

4.1 The Problem Statement

The problem is to design an architecture that satisfies the core Kanban requirements while enforcing separation of concerns and strict dependency rules. The Layered Architecture pattern is chosen as the first structural approach to address the Modifiability, Security, Reliability, Scalability, and Performance ASRs identified in Lab 1.

4.2 Impact of ASRs on Layered Architecture

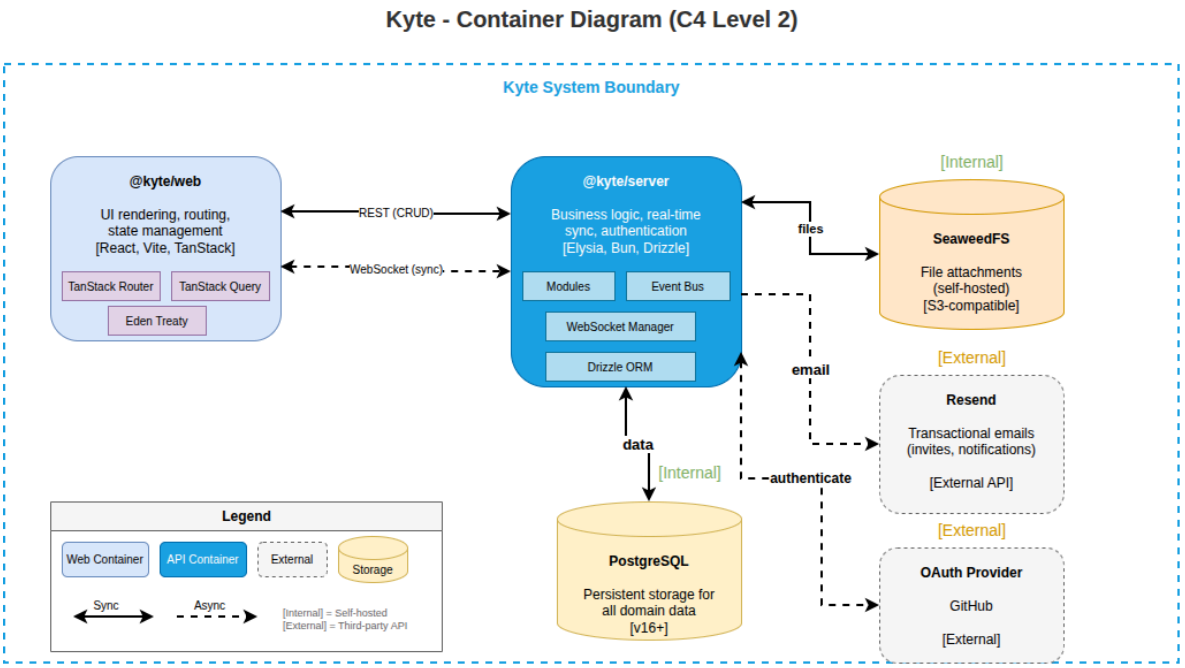
- ASR-1 (Modifiability) -> Layered Structure: ASR-1 demands that changes to task types or external integrations remain isolated. The Layered Architecture inherently supports this by strictly defining four layers (Presentation, Business Logic, Persistence, Database) with downward dependencies.
  - This means: The Business Logic (where task movement rules reside) can be modified without impacting the Presentation layer (UI). This also enables future Mobile App development using the same backend services.
- ASR-2 (Security) -> Business Logic Layer: ASR-2 requires rigorous authorization for board modifications.
  - In the Layered Architecture, this logic is enforced in the Business Logic Layer through a dedicated Security middleware.
  - This ensures that security checks are applied to the core services (e.g., TaskService, BoardService), protecting them from unauthorized access

regardless of whether the request originates from the Web App or a Mobile API.

- ASR-3 (Reliability) -> Persistence Layer & Business Logic Layer: ASR-3 requires handling concurrent task modifications without data loss.
  - The Persistence Layer must implement version control or timestamp fields on Task entities to detect conflicts.
  - The Business Logic Layer must implement conflict detection and resolution mechanisms (optimistic locking) to handle simultaneous updates safely.
  - This means: When two users move the same task simultaneously, the system will detect the conflict and resolve it without corrupting data.
- ASR-4 (Scalability) -> Presentation Layer: ASR-4 requires stateless API design to enable horizontal scaling.
  - The Presentation Layer must avoid storing server-side local sessions, using session cookies (via Better Auth) with data in PostgreSQL instead.
  - RESTful API patterns are enforced to support growth from 100 to 1,000 concurrent active boards through load balancing.
  - This means: Multiple server instances can handle requests without session synchronization overhead.
- ASR-5 (Performance) -> Persistence Layer & Business Logic Layer: ASR-5 requires API response times under 300ms.
  - The Persistence Layer must implement proper database indexing strategies on frequently queried fields (e.g., board_id, column_id).
  - The Business Logic Layer may implement caching mechanisms for frequently accessed data (e.g., board metadata, user permissions).
  - This means: Database queries are optimized and repeated data fetches are minimized through caching.

The next step (Lab 2) will use these requirements to formally define the four architectural layers and model the logical components (Controller, Service, Repository) and their interfaces within those layers.

4.3 Layered Architecture Overview



5. Conclusion & Reflection

The requirements elicitation phase for Kyte successfully defined the project scope through detailed Functional, Non-Functional, and Architecturally Significant Requirements. The UML Use Case Diagram provides a clear visual model of user interactions and the critical path. The identified ASRs especially those relating to Modifiability and Security directly necessitate the adoption of a structured pattern like the Layered Architecture for the subsequent design phase (next Lab 2), ensuring the architecture can sustainably support the required system behaviors.